CHEMISTRY PROJECT

SUBMITTED BY
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XII B

Acknowledgement

I take this opportunity to thank our Principal, Fr. Joice Elavathingal, Academic Administrator Mr. Atly T. John, and Senior Secondary Co-ordinator Ms. Suma for their support and encouragement. I would also like to express my sincere gratitude to all the teachers of the Chemistry Department who guided me to the successful completion of this project. I am indebted to my parents and classmates, who helped me a lot in completing the project within the limited time frame.

Introduction

Fruits are naturally rich in essential nutrients, particularly simple sugars such as glucose and fructose. These sugars are classified as reducing sugars because they can donate electrons to other molecules during chemical reactions. In today's dietary habits, fruit juices—both freshly made and commercially packaged—are consumed on a large scale. However, the sugar level in these juices can differ based on the type of fruit, method of preparation, and the addition of sweeteners or preservatives. For this reason, it becomes important to measure the sugar content scientifically to evaluate its nutritional value and potential health effects.

The objective of this project is to determine the sugar content in various fruit juices using a classical chemical approach that involves Fehling's solution. This technique works on the principle of a redox reaction, where reducing sugars react with copper(II) ions in Fehling's solution to produce a reddish precipitate of copper(I) oxide. The extent of this reaction can be used to estimate the amount of reducing sugar present in the sample.

Fehling's solution itself is prepared by mixing two separate components: Fehling's solution A, which contains copper(II) sulphate, and Fehling's solution B, which contains potassium sodium tartrate and sodium hydroxide. When these two are combined in equal volumes, they form Fehling's reagent, which specifically reacts with reducing sugars like glucose, making it suitable for sugar estimation.

In the experiment, a standard glucose solution is first prepared and used to standardise the Fehling's reagent. This helps determine the exact amount of glucose needed to reduce all the copper(II) ions to copper(I) oxide. The reaction is accompanied by a change in colour from blue to red, with methylene blue used as an indicator to identify the endpoint of titration.

After standardisation, the same procedure is applied to different fruit juice samples. A measured volume of juice is diluted and titrated with Fehling's reagent. The reducing sugars in the juice react in the same way, allowing the sugar concentration of the juice to be calculated using the earlier standardisation results.

This method is not only reliable and accurate but also provides hands-on experience with titration, redox reactions, and the chemistry of carbohydrates. It also raises

awareness about the sugar hidden in everyday foods and highlights how chemistry is applied in real life for food testing, quality control, and health-related studies.

Results and Analysis

From this analysis, it is evident that while apple and orange juices contain moderate levels of glucose, grape juice has the lowest glucose concentration of the three. Consumers seeking to manage their glucose intake may prefer grape juice, while those looking for a balance between flavour and sweetness may favour apple or orange juice. Additionally, those needing an energy boost from glucose may find orange juice more suitable due to its higher glucose content compared to apple and grape juice.

This report covers the glucose concentration findings for the three juice samples, offering both scientific data and practical consumer consideration.

Conclusion

The analysis of sugar content in various fruit juices revealed significant differences across brands and types of juices. Some juices, particularly those marketed as “all natural,” had surprisingly high sugar levels. When compared to the World Health Organization’s guidelines on daily sugar intake, many of the tested juices contributed a large portion of the recommended limit, raising concern about their potential impact on health. Regular consumption of high-sugar juices can increase the risk of health issues such as obesity, type 2 diabetes, and dental problems. Despite being perceived as healthier alternatives to sodas, some fruit juices contain sugar levels that are just as high, if not higher. This finding emphasizes the importance of reading nutritional labels and opting for lower-sugar or diluted juice options to manage sugar intake efficiently.